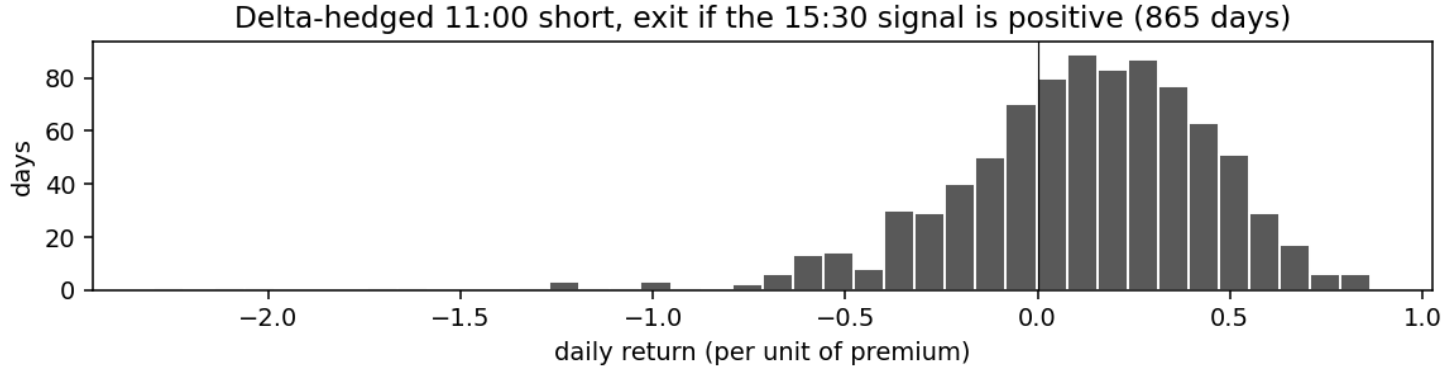


0DTE delta-hedged ATM straddles, 11:00 entry held till close/15:30

Every day sells the nearest-OTM straddle at 11:00 and delta-hedges to 16:00. At 15:30 each forecast model decides on its own whether to buy the straddle back: exit the short if *that* model's $s > 0$ (its realized variance forecast above implied); otherwise cash-settle.

	n	mean	std	min	max	t	Sharpe _{ann}	Sharpe _{ann} [×]	MaxDD	MaxDD [×]	n_{exit}	%exit
hold through cash-settle	865	0.103	0.388	-2.30	0.87	7.78	4.20	3.58	-3.48	-3.80	0	0.0
exit if $s > 0$												
baseline (HAR + calendar OLS)	865	0.102	0.377	-2.30	0.87	7.96	4.30	3.66	-3.21	-3.57	286	33.1
block-diagonal ridge	865	0.104	0.365	-2.30	0.87	8.36	4.51	3.85	-3.84	-4.25	346	40.0
block-diagonal ridge, without the FOMC columns	865	0.106	0.357	-2.30	0.87	8.74	4.72	4.05	-3.81	-4.21	355	41.0
LightGBM	865	0.108	0.370	-2.30	0.87	8.62	4.65	4.00	-3.38	-3.78	289	33.4
XGBoost	865	0.108	0.370	-2.30	0.87	8.56	4.62	3.97	-3.75	-4.15	278	32.1
lasso (causally tuned)	865	0.107	0.366	-2.30	0.87	8.64	4.66	4.01	-3.53	-3.94	321	37.1
lasso (fixed 1e-4)	865	0.103	0.365	-2.30	0.87	8.33	4.50	3.84	-3.81	-4.21	358	41.4
elastic net (causally tuned)	865	0.105	0.367	-2.30	0.87	8.38	4.52	3.87	-3.85	-4.25	307	35.5

Delta hedge. (1) Sell the nearest-OTM call and put at 11:00; those strikes stay unchanged. (2) Every later 30-minute stamp until 15:30, read the vendor spot and vendor hourly IV at that stamp. Total vol is that IV times the square root of hours remaining to 16:00. Straddle delta is Black-76 call delta plus put delta (interest rate zero, forward = spot). Because the straddle is short, hold plus-delta of the index. Stock P&L over the next half hour is delta times the spot change. (3) At 15:30, if *that row's* model has $s > 0$, buy the straddle back at the Black-76 mark and close the stock. If not, cash-settle versus the GSPC close (last stock step uses that close versus the 15:30 spot). No look-ahead: the entry uses only the 11:00 quote; each rebalance uses only the quote at that stamp.



Daily return of the 11:00 short; 15:30 exit uses the block-diagonal ridge s .

One expiration day is one return. $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$. t -stat is $\sqrt{n} \cdot \text{mean}/\text{std}$. Every day enters at 11:00. Each model only decides whether to exit at 15:30. n_{exit} is days that model buys the straddle back. $\text{Sharpe}_{\text{ann}}^{\times}$ is the same book at the crossed spread: sell the 11:00 straddle at the bid, then cash-settle or buy back at the 15:30 mark; P&L is still divided by the midpoint premium. MaxDD is the worst peak-to-trough decline of the cumulative sum of the daily returns, in date order, one unit a day (summed, not compounded), signed and in units of entry premium like min; MaxDD[×] is the same at the crossed spread.